Corentin Léger - Machine Learning Research Engineer
Website, GitHub, LinkedIn, Google Scholar

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Research Engineer with 3 years of experience in Reinforcement Learning, Large Language Models, Distributed computing and Multi-Agent systems across academic and industry settings. Interested in RL, LLMs and open-ended learning.

Skills

- **Technical**: Python, PyTorch, Distributed Systems (Ray), Scikit-Learn, JAX, Optuna, Hydra, vLLM, CI/CD
- **Research**: Machine Learning, Reinforcement Learning, AutoML, Large Language Models

Experience

- **InstaDeep** — Paris, France
  Research Engineer — *Oct 2025 - Present*
  - Reinforcement Learning research for real-world train dispatching optimization with Deutsche Bahn.
  - Designed and implemented a parallelized hyperparameter optimization pipeline (Ray Tune), achieving a 4× speedup over the previous workflow. It is now used by a team of 10 researchers.
  - Reduced mean train delays in eval by 50% across single and multi-agent approaches without degrading other target metrics, by introducing a dense reward shaping mechanism as well as tuning reward and PPO parameters.
  - Implemented attention-based encoders for multi-train and route observations, and improved PPO critic input representation, improving both sample efficiency and final policy performance metrics by about 5%.
  - Discovered and fixed a silent data-loss bug in our distributed rollout sampling stack (RLlib). Improved experiment pipelines robustness, and code quality through code reviews and automated tests for CI/CD.

- **Huawei - Noah's Ark Lab** — Paris, France
  Research Engineer — *Jan 2025 - Oct 2025*
  - Automated Machine Learning and LLMs research at Paris Noah's Ark Lab, supervised by Balázs Kégl.
  - Developed a tabular data preprocessing pipeline combining feature engineering, with LLMs code generation (vLLM) and heuristics, and feature selection methods to enhance the performance of our AutoML tool.
  - Created a post-training pipeline with TRL to fine-tune Llama3 8B on feature engineering code generation with GRPO.
  - Contributed to papers studying LLM hyperparameter generation skills (NeurIPS 2025 workshop) and bilevel optimization for Reinforcement Learning (preprint).

- **Inria - Flowers Lab** — Bordeaux, France
  Research Engineer — *Dec 2023 - Dec 2024*
  - LLMs and Multi-agent systems research at Flowers Lab, supervised by Clément Moulin Frier.
  - Contributed to two papers (ICLR 2025, preprint) studying text evolution in multi-turn LLM interactions. Built a multi-agent simulator, NLP and data visualization tools to run and analyze experiments (LLM-Culture).
  - Co-developed a framework enabling evolutionary prompt optimization for multi-agent debate (NeurIPS 2025 workshop).
  - Co-developed Vivarium, a high performance multi-agent simulator built in Jax, that can be interacted with in real-time from a web interface or a Jupyter notebook. Used it to teach a Master's course at UPF Barcelona.

  Research Scientist Intern — *May 2023 - Nov 2023*
  - Meta-Reinforcement Learning research at Flowers Lab, supervised by Clément Moulin Frier
  - Published a method (EvoAPPS 2024) improving Deep RL agent's abilities by optimizing RNNs with a distributed evolutionary search (SLURM, Optuna). It enables to solve partially observable tasks, and adapt faster to unseen environments.
  - Co-authored a paper studying the combination of Tabular Reinforcement Learning and Symbolic data (HAL Inria).

- **Connectiv-IT** — Bordeaux, France
  Data Scientist Intern — *May 2022 - Aug 2022*
  - Preprocessed helicopter maintenance data, performing cleaning, outlier detection and imputation (Pandas, Scikit-Learn). Used statistical analysis (SciPy) and clustering to identify key trends in maintenance data.

Education

- **Ecole Nationale Supérieure de Cognitique** — Bordeaux, France
  Master of Science in Computer and Cognitive Sciences — *Sept. 2020 – Sept. 2023*
  Activities: Bronze medal at French University Volley-Ball Championship 2023

- **Cycle Préparatoire de Bordeaux** — Bordeaux, France
  Bachelor of Science in Mathematics and Physics, Sport-Study contract with ASI Volley-ball — *Sept. 2018 – Jun. 2020*